Nutrient Cycles

Exam code: 0610 & 0970

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The Carbon Cycle

- Nutrients such as carbon and nitrogen are not endless resources
- There is a finite amount of each element on the planet and as such, they need to be **recycled** in order to allow new organisms to be made and grow
- **Carbon** is taken out of the atmosphere in the form of **carbon dioxide** by plants to be used for photosynthesis
- It is passed on to animals (and microorganisms) by **feeding**
- It is returned to the atmosphere in the form of **carbon dioxide** by plants, animals and microorganisms as a result of **respiration**
- If animals and plants die in conditions where decomposing microorganisms are not present the carbon in their bodies can be converted, over millions of years and significant pressure, into **fossil fuels**
- When fossil fuels are burned (the process is known as **combustion**), the carbon combines with oxygen and **carbon dioxide is released** into the atmosphere
- **Increased use of fossil fuels** is contributing to an increase in the carbon dioxide content of the atmosphere
- In addition, **mass deforestation** is **reducing the amount of producers** available to take carbon dioxide out of the atmosphere by photosynthesis
- This problem is exacerbated by the fact that in many areas of the world, deforestation is taking place for land rather

Help me understand

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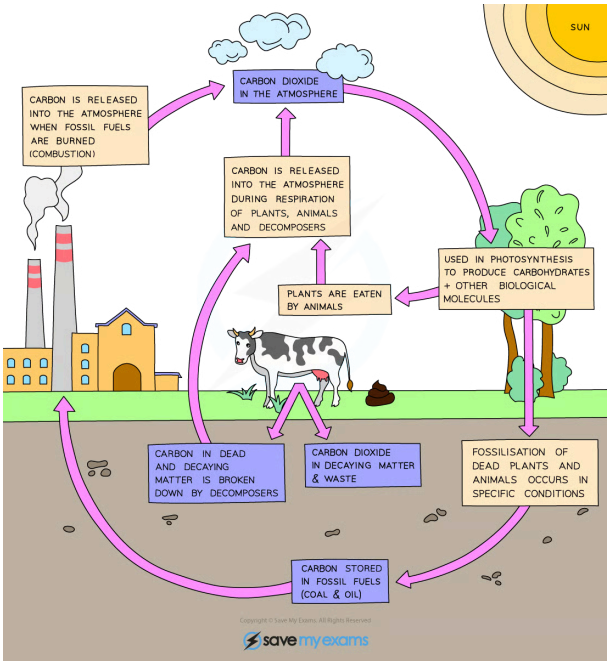
Got it



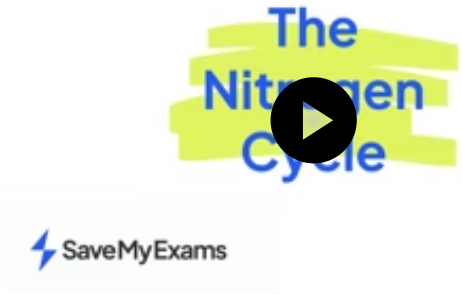
The Carbon Cycle

Examiner Tips and Tricks

The carbon cycle is simple:

- Carbon is taken out of the atmosphere by photosynthesis
- It is passed on to animals and decomposers by feeding
- It is returned by respiration; in plants, in animals and in decomposing
- microorganisms
  In addition, it is returned (in increasing amounts) by combustion of fossil fuels

You should be able to identify what each arrow represents in any diagram of the carbon cycle.



The Nitrogen Cycle: Extended

- **Nitrogen** as an element is required to make proteins
- Neither plants nor animals can **absorb it from the air** as N_2 gas is very stable and cannot be easily broken down
- Two ways that Nitrogen gas in the air can be converted into a usable form are:
  - **Nitrogen fixing bacteria** convert N_2 gas into ammonium compounds, which can then be converted to usable **nitrates**
    - Nitrogen fixing bacteria can be free-living in the soil or can live within the root nodules of some plants
  - **Lightning** can split the bond between the two N atoms, turning them into nitrous oxide N_2O and nitrous dioxide NO_2 that dissolve in rainwater and leach into the soil
- Plants absorb nitrogen from the soil in the form of **nitrates** and use it to build proteins
- Animals **eat** the plants and get the nitrogen they need from the plant proteins
  - This nitrogen is **passed up the food chain** when secondary consumers eat primary consumers, etc.
- **Waste** (urine and faeces) from animals sends nitrogen back into the soil in the form of ammonium compounds
  - E.g. the urea in urine contains nitrogen
- The bodies of dead plants and animals **decay** and all the proteins inside them are broken down into ammonium compounds by decomposers
- The plants can't absorb ammonium compounds, so **nitrifying bacteria** convert the ammonium compounds to nitrites and then to nitrates, which can then be absorbed by plants – and so the cycle goes on
- **Denitrifying bacteria** take nitrates **out of the soil** and convert them back into N_2 gas
  - This process **reduces soil fertility** and is bad for plant growth
  - Denitrifying bacteria are anaerobic so aerating the soil, e.g. by reducing waterlogging and turning over the soil during ploughing, can reduce the rate of denitrification

The nitrogen cycle involves nitrogen fixation, decomposition, nitrification and denitrification

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